

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO2:** Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Industry Problem Task

Duration: 2.5hrs

1. Explain how Reinforcement Learning can be applied to optimize delivery routes in a logistics system. Identify the possible states, actions, rewards, and policy. Discuss how continuous learning improves efficiency over time. **(5 Marks)**
2. Apply Reinforcement Learning concepts to a fraud detection system in fintech. Define the state space, action space, and reward mechanism. Explain how exploration strategies improve detection accuracy. **(5 Marks)**
3. Illustrate how Reinforcement Learning is used in a streaming platform for content recommendation. Discuss the impact of delayed rewards, user feedback, and changing preferences on learning performance. **(5 Marks)**

4. Evaluate the effectiveness of Reinforcement Learning in predictive maintenance compared to traditional rule-based approaches. Discuss risks and reliability concerns in industrial environments. **(5 Marks)**

5. Analyze how Reinforcement Learning can be used to optimize transportation systems such as bus scheduling or route allocation. Define states, actions, and rewards, and discuss how real-time data improves performance. **(5 Marks)**

6. Apply Reinforcement Learning to dynamic pricing in e-commerce. Define the state space, action space, and reward function. Discuss challenges in balancing exploration and exploitation. **(5 Marks)**

7. Illustrate the use of Reinforcement Learning in smart city waste management systems. Identify the MDP components and discuss how continuous updates improve operational efficiency. **(5 Marks)**

8. Evaluate how Reinforcement Learning can be used in network bandwidth allocation in telecommunications. Define states, actions, and rewards, and examine how the system adapts to dynamic conditions. **(5 Marks)**

9. Analyze the application of Reinforcement Learning in portfolio management in finance. Discuss advantages over traditional models and challenges such as risk and delayed rewards. **(5 Marks)**

10. Justify the use of Reinforcement Learning in renewable energy management systems. Define the RL framework components and discuss the impact of environmental uncertainty on decision-making. **(5 Marks)**

1. Delivery Route Optimization (Logistics)

Objective

Minimize delivery time, fuel cost, and delays.

Why RL?

Environment is dynamic (traffic, weather). RL adapts through experience.

MDP Components

- **States:** Vehicle location, traffic, pending deliveries, fuel level
- **Actions:** Choose next route, stop, reroute
- **Environment:** Road network with real-time conditions

Reward Function

- +10 on successful delivery
- -1 per time step (delay penalty)
- -5 for long routes
- -20 for failed delivery

Policy

Maps state → best route decision

Learning Process

- Uses **Q-learning / Deep RL**
- Updates using reward feedback
- Improves via trial-and-error

Continuous Learning

Adapts to traffic patterns → faster deliveries over time

Flow (RL Architecture)

State → Action → Environment → Reward → Update Policy → Repeat

Real Example

Amazon delivery route optimization

Advantages

- Adaptive routing
- Reduced cost & time
- Real-time decision making

2. Fraud Detection (FinTech)

Objective

Detect fraudulent transactions accurately.

Why RL?

Fraud patterns evolve → RL adapts continuously.

MDP Components

- **States:** Transaction amount, location, user history
- **Actions:** Approve, flag, block

Reward Function

- +10 correct fraud detection
- -20 missed fraud
- -5 false alarm

Policy

Chooses action maximizing detection accuracy

Exploration Strategy

- ϵ -greedy: tries new patterns to detect unseen fraud

Learning Process

- Updates based on feedback (fraud confirmed later)

Continuous Learning

Improves detection accuracy with new fraud patterns

Flow

Transaction → Decision → Feedback → Policy Update

Real Example

PayPal fraud detection system

Advantages

- Detects new fraud types
- Reduces financial loss
- Improves accuracy over time

3. Streaming Recommendation System

Objective

Maximize user engagement and watch time

Why RL?

User preferences change dynamically

MDP Components

- **States:** User history, preferences, time of day
- **Actions:** Recommend movie/content

Reward Function

- +5 watch
- +10 completion
- -3 skip

Delayed Rewards

User satisfaction seen later → affects learning

Policy

Suggests content maximizing engagement

Learning Process

- Learns from clicks, watch time
- Uses Deep RL

Continuous Learning

Adapts to changing user tastes

Flow

User → Recommendation → Feedback → Update

Real Example

Netflix recommendation system

Advantages

- Personalized content
- Increased engagement
- Better user retention

4. Predictive Maintenance (Industry)

Objective

Prevent machine failures and reduce downtime

Why RL?

Handles complex, uncertain machine behavior

MDP Components

- **States:** Machine condition, vibration, temperature
- **Actions:** Repair, replace, continue operation

Reward Function

- +10 prevent failure
- -15 unexpected breakdown
- -5 maintenance cost

Policy

Decides optimal maintenance timing

Learning Process

- Learns patterns of failure
- Updates from machine data

Continuous Learning

Improves predictions with more data

Comparison with Rule-Based

RL	Rule-Based
Adaptive	Static
Learns patterns	Fixed thresholds
Better performance	Less flexible

Risks

- Wrong decision → costly failures
- Needs high-quality data

Real Example

GE industrial predictive maintenance

Advantages

- Reduced downtime
- Cost savings
- Increased reliability

5. Dynamic Pricing (E-commerce)

Objective

Maximize profit while maintaining demand

Why RL?

Market demand changes frequently

MDP Components

- **States:** Demand, competitor price, inventory
- **Actions:** Increase/decrease price

Reward Function

- Profit = revenue - cost
- Penalize loss of customers

Policy

Chooses optimal price dynamically

Exploration vs Exploitation

- Explore: try new prices
- Exploit: use best-known price

Learning Process

- Learns demand-price relationship

Continuous Learning

Adapts to market trends

Flow

State → Price → Sales → Reward → Update

Real Example

Amazon dynamic pricing

Advantages

- Maximized revenue
- Competitive pricing
- Real-time adaptation

Code of Conduct:

*I **E. V. Mahendra Reddy(192425214)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts

Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation